Parsa Esmati

PhD candidate in machine learning, University of Bristol

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PROFILE

Machine learning researcher and expert in **representation learning and visual generation**, across **diffusion, flow and discrete models**. Lead author of **three NeurIPS papers, including one Spotlight** (top 3%), with experience pretraining **large-scale video world models at Microsoft Research** and hands-on expertise in **multi-node distributed training**. Trains, adapts and optimises **frontier visual generative models and architectures** for both training and inference at scale.

RESEARCH EXPERIENCE

Research Intern · **Microsoft Research**, Game Intelligence Group — Jun 2025 – Oct 2025
Managers: Katja Hofmann, Raluca Stevenson

- **Pretrained the tokenizer and the generator** of an action-conditioned video world model system, at 100-GPU scale with FSDP and DDP through DeepSpeed and PyTorch Lightning.
- Developed **Latent Anchored Binary Quantization**, a discrete visual tokenizer targeting straight-through-estimator error; trained tokenizers and discrete image generators at 32-GPU scale (under review, ICLR 2027).

Doctoral Researcher · Machine Learning and Computer Vision Group, **University of Bristol** — Sep 2022 – present
Advisor: Prof. Majid Mirmehdi

- **Physical reasoning in video diffusion.** Built a reverse sampling strategy to unravel the physical structure video diffusion models encode but do not render (**NeurIPS 2026 Spotlight**, top 3%, with Microsoft Research, Mila, McGill, Calgary).
- **Guided flow sampling.** Adapted large-scale visual generative pipelines and improved their sampling with training-free, probability-conserving guidance for flow and diffusion models in image generation (**NeurIPS 2026**, with KAIST).
- **Latent geometry and generative difficulty.** The Nesting Hypothesis: denoising-optimal latent manifolds and information nesting in representation encoders; image generative models trained at 32-GPU scale (preprint).
- **Curriculum self-training.** Source-free video domain adaptation with collaborative curriculum self-training and VLM adaptation (**WACV 2025**).

SELECTED PUBLICATIONS

[1] **P. Esmati**, S. Nath, K. Hofmann, D. Nowrouzezahrai, S. E. Kahou, M. Mirmehdi. *The Invisible Hand of Physics: When Video Diffusion Models Know More Than They Show.* arXiv:2606.05328 — **NeurIPS 2026** ★ **Spotlight** · top 3%

[2] **P. Esmati**, J. Hyung, A. Dadashzadeh, J. Choo, M. Mirmehdi. *Probability-Conserving Flow Guidance.* arXiv:2605.20079 — **NeurIPS 2026**

[3] **P. Esmati**, M. Mirmehdi. *The Nesting Hypothesis: How Latent Geometry Shapes Generative Difficulty.* — Preprint

[4] **P. Esmati**, J. Hyung, E. Aiello, R. Stevenson, T. He, J. Bian, M. Mirmehdi, K. Hofmann. *Visual Tokenization with Latent Anchored Binary Quantization.* — Under review, ICLR 2027

[5] **P. Esmati**, A. Dadashzadeh, V. G. Ardakani, N. Larrosa, N. Grilli. *SEA: State-Exchange Attention for High-Fidelity Physics Based Transformers.* — **NeurIPS 2024**

[6] A. Dadashzadeh, **P. Esmati**, M. Mirmehdi. *Co-STAR: Collaborative Curriculum Self-Training with Adaptive Regularization for Source-Free Video Domain Adaptation.* — **WACV 2025**

ENGINEERING & LARGE-SCALE TRAINING

Languages	Python, C++, CUDA, C, MATLAB.
Frameworks	PyTorch, DeepSpeed, PyTorch Lightning, libtorch; TensorFlow; JAX (basic).
Distributed training & inference	DDP and FSDP with DeepSpeed and Lightning; sequence parallelism. Training runs on 100 GPUs (video world models, Microsoft Research) and 32 GPUs (image generative models and visual tokenizers).
Infrastructure & tooling	SLURM, Docker, GitHub, CI/CD pipelines.
Environments	Microsoft Azure; Isambard-AI Phase 2 (UK national AI supercomputer, granted allocation).

EDUCATION

PhD in Engineering (Machine Learning) · **University of Bristol** — Sep 2022 – Nov 2026 (expected)
Thesis: *Representation Geometry and Dynamics for Efficient Latent Generation.* Advisor: Prof. Majid Mirmehdi.
Examiners: Prof. Björn Ommer, Dr. Mengyue Yang.
Prior to this, numerical solver development for PDEs in C++, leading to State-Exchange Attention (NeurIPS 2024).

Bachelor's and Master's in Engineering · **University of Bristol** · Control theory, robotics and computational methods — 2018 – 2022

SERVICE & TEACHING

Reviewer	NeurIPS, ICLR, ICML.
Teaching, University of Bristol	Python programming, C++ programming, engineering sciences, fluid mechanics and heat transfer, aerospace vehicle design and system integration.